# CSTR DRTO Formulation: Continuous Model, Discretization, and RL-DRTO Architecture

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## Contents

# 1 Overview

We propose a two-layer architecture for reinforcement-learning-based real-time optimization (RL-RTO) of an exothermic continuous stirred tank reactor (CSTR) with a cooling jacket and PI feedback control. The structure separates:

(a) the **upper layer**: a Dynamic Real-Time Optimization (DRTO) or its RL replacement (RL-DRTO), which computes economic setpoints for the reactor temperature; and

(b) the **lower layer**: the PI-controlled reactor environment representing the true plant dynamics.

The DRTO layer may include simplified dynamics for prediction, while the environment includes the plant dynamics and controller. This separation enables real-time deployability.

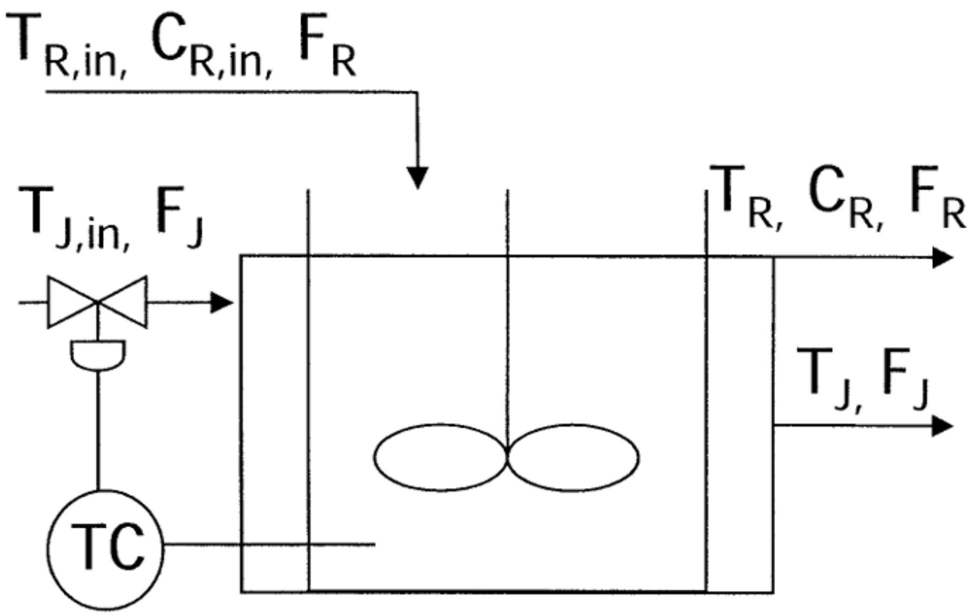


Figure 1: Illustration of the CSTR case.

# 2 Notation and Parameters

## 2.1 Time horizon and indices

Continuous time horizon: $t \in [0, t_f]$.
Discrete grid (used in Section 5): $t_k = k\,\Delta t$, $k = 0, 1, \ldots, N$ with $\Delta t = t_f/N$.

## 2.2 Geometry

Reactor volume and area (function of design parameters $D_R$ and $H_R$):

$$V_R = \frac{\pi}{4} D_R^2 H_R, \qquad\qquad A_R = \pi D_R H_R. \tag{1}$$

### 2.3 Model variables and parameters

Table 1: Decision variables (continuous-time).

| Symbol | Meaning | Bounds / Units |
|---|---|---|
| $C_R(t)$ | Reactant concentration in reactor | $0 \le C_R \le C_{R,in}$ (mol A/m$^3$) |
| $T_R(t)$ | Reactor temperature | $T_{R,\min} \le T_R \le T_{R,\max}$ (K) |
| $T_J(t)$ | Jacket temperature | $T_{J,\min} \le T_J \le T_{J,\max}$ (K) |
| $F_J(t)$ | Jacket flow rate (control) | $F_{J,\min} \le F_J \le F_{J,\max}$ (m$^3$/hr) |
| $s_{\text{conv}}(t)$ | Slack for conversion constraint | $s_{\text{conv}} \ge 0$ |
| $s_{\text{cool}}(t)$ | Slack for cooling/heat duty constraint | $s_{\text{cool}} \ge 0$ |
| $s_{\text{temp}}(t)$ | Slack for temperature gradient constraint | $s_{\text{temp}} \ge 0$ |

Table 2: Model parameters.

| Symbol | Meaning | Units |
|---|---|---|
| $D_R$ | Reactor diameter (design) | m |
| $H_R$ | Reactor height (design) | m |
| $k_0$ | kinetic rate constant | hr$^{-1}$ |
| $E_a/R$ | Activation energy | K |
| $\Delta H$ | Heat of reaction | KJ/mol |
| $\rho, C_p$ | Reactor liquid density, heat capacity | kg/m$^3$, kJ/kg K |
| $\rho_J, C_{p,J}$ | Jacket liquid density, heat capacity | kg/m$^3$, kJ/kg K |
| $U, A_R$ | Overall heat transfer coeff. and area | kJ/hr m$^2$ K, m$^2$ |
| $V_R(V_J)$ | Reactor(Jacket) volume | m$^3$ |
| $F_R(F_J)$ | Reactor(Jacket) feed volumetric flow rate | m$^3$/hr |
| $C_{R,in}$ | Inlet reactant concentration | mol A/m$^3$ |
| $T_{R,in}, T_{J,in}$ | Inlet temperatures (reactor feed, jacket) | K |
| *Operational parameters* | | |
| $T_{R,\min}, T_{R,\max}$ | Reactor temperature bounds | K |
| $T_{J,\min}, T_{J,\max}$ | Jacket temperature bounds | K |
| $F_{J,\min}, F_{J,\max}$ | Jacket flow bounds | m$^3$ h$^{-1}$ |
| $C_{R,0}, T_{R,0}, T_{J,0}$ | Initial concentration and temperatures | mol A/m$^3$ |
| $X_{\min}$ | Minimum required conversion | – |
| $Q_{\max}$ | Max cooling/heat duty allowance | W |
| $\Delta T_{\max}$ | Max allowed temperature change rate | K/hr |
| $P_{\text{prod}}$ | Product price | \$/mol |
| $P_{\text{react}}$ | Reactant cost | \$/mol |
| $P_{\text{energy}}$ | Energy cost (cooling/heating) | \$/kJ |
| $w_{\text{conv}}$ | Penalty weight for conversion slack | \$/(unit) |
| $w_{\text{cool}}$ | Penalty weight for cooling slack | \$/(unit) |
| $w_{\text{temp}}$ | Penalty weight for temperature ramp slack | \$/(unit) |

# 3 Continuous-Time Model

## 3.1 Reaction kinetics

$$k(t) = k_0 \exp\left(-\frac{E_a}{R\,T_R(t)}\right). \tag{2}$$

## 3.2 Mass balance

$$\frac{dC_R}{dt} = \frac{F_R}{V_R}\big(C_{R,in} - C_R\big) - k\,C_R. \tag{3}$$

## 3.3 Energy balances

**Reactor.**

$$\frac{dT_R}{dt} = \frac{F_R}{V_R}\big(T_{R,in} - T_R\big) - \frac{\Delta H}{\rho C_p}\,k\,C_R - \frac{UA_R}{\rho C_p V_R}\big(T_R - T_J\big). \tag{4}$$

**Jacket.**

$$\frac{dT_J}{dt} = \frac{F_J}{V_J}\big(T_{J,in} - T_J\big) + \frac{UA_R}{\rho_J C_{p,J} V_J}\big(T_R - T_J\big). \tag{5}$$

# 4 Operational Constraints (Continuous-time)

**Box bounds.** As listed in Table 1 and Table 2.

**Temperature change (soft).** The absolute change is limited with a nonnegative slack:

$$\left|\frac{dT_R}{dt}\right| \le \Delta T_{\max} + s_{\text{temp}}(t), \qquad s_{\text{temp}}(t) \ge 0. \tag{6}$$

**Conversion (soft).**

$$C_{R,in} - C_R(t) + s_{\text{conv}}(t) \ge X_{\min}, \qquad s_{\text{conv}}(t) \ge 0. \tag{7}$$

**Cooling/heat duty (soft).**

$$UA_R\,\big|T_R(t) - T_J(t)\big| \le Q_{\max} + s_{\text{cool}}(t), \qquad s_{\text{cool}}(t) \ge 0. \tag{8}$$

**Initial conditions.** can be set from the environment/plant measurements.

$$C_R(0) = C_{R,0}, \quad T_R(0) = T_{R,0}, \quad T_J(0) = T_{J,0}. \tag{9}$$

# 5 Backward-Euler Discretization and Discrete Model

Let $k = 1, \ldots, N$ with $t_k = k\,\Delta t$. Denote $C_{R,k} \approx C_R(t_k)$ (similarly for $T_{R,k}$, $T_{J,k}$, slacks and controls). Backward–Euler for any state $x$ reads

$$\left.\frac{dx}{dt}\right|_{t=t_k} \approx \frac{x_k - x_{k-1}}{\Delta t}. \tag{10}$$

### 5.1 Discrete variables (per time step $k$)

States: $C_{R,k}$, $T_{R,k}$, $T_{J,k}$; Manipulated variable: $F_{J,k}$; Slacks: $s_{\mathrm{conv},k}$, $s_{\mathrm{cool},k}$, $s_{\mathrm{temp},k}$.

### 5.2 Discrete kinetics

$$k_k = k_0 \exp\left(-\frac{E_a}{R\,T_{R,k}}\right). \tag{11}$$

### 5.3 Discrete dynamics (Backward–Euler)

$$C_{R,k} = C_{R,k-1} + \Delta t \left[\frac{F_R}{V_R}(C_{R,in} - C_{R,k}) - k_k C_{R,k}\right], \tag{12}$$

$$T_{R,k} = T_{R,k-1} + \Delta t \Big[\frac{F_R}{V_R}(T_{R,in} - T_{R,k}) - \frac{\Delta H}{\rho C_p} k_k C_{R,k} - \frac{UA_R}{\rho C_p V_R}(T_{R,k} - T_{J,k})\Big], \tag{13}$$

$$T_{J,k} = T_{J,k-1} + \Delta t \Big[\frac{F_{J,k}}{V_J}(T_{J,in} - T_{J,k}) + \frac{UA_R}{\rho_J C_{p,J} V_J}(T_{R,k} - T_{J,k})\Big]. \tag{14}$$

### 5.4 Discrete constraints

**bounds.**

$$0 \le C_{R,k} \le C_{R,in}, \qquad T_{R,\min} \le T_{R,k} \le T_{R,\max}, \qquad T_{J,\min} \le T_{J,k} \le T_{J,\max}, \tag{15}$$

$$F_{J,\min} \le F_{J,k} \le F_{J,\max}, \qquad s_{\mathrm{conv},k}, s_{\mathrm{cool},k}, s_{\mathrm{temp},k} \ge 0. \tag{16}$$

**Temperature ramp (soft).**

$$-\Delta T_{\max} - s_{\mathrm{temp},k} \;\le\; T_{R,k} - T_{R,k-1} \;\le\; \Delta T_{\max} + s_{\mathrm{temp},k}. \tag{17}$$

**Conversion (soft).**

$$C_{R,in} - C_{R,k} + s_{\mathrm{conv},k} \ge X_{\min}. \tag{18}$$

**Cooling/heat duty (soft).**

$$UA_R\,|T_{R,k} - T_{J,k}| \le Q_{\max} + s_{\mathrm{cool},k}. \tag{19}$$

**Initial conditions.**

$$C_{R,0} = C_{R,0}^{\mathrm{given}}, \quad T_{R,0} = T_{R,0}^{\mathrm{given}}, \quad T_{J,0} = T_{J,0}^{\mathrm{given}}. \tag{20}$$

### 5.5 Discrete economic objective

$$\begin{aligned}\max \sum_{k=1}^{N} \Delta t \Big(P_{\mathrm{prod}}\, F_R\,(C_{R,in} - C_{R,k}) - P_{\mathrm{energy}}\, UA_R\,|T_{R,k} - T_{J,k}|\Big) \\ - \sum_{k=1}^{N} \Delta t \left(w_{\mathrm{conv}}\, s_{\mathrm{conv},k} + w_{\mathrm{cool}}\, s_{\mathrm{cool},k} + w_{\mathrm{temp}}\, s_{\mathrm{temp},k}\right).\end{aligned} \tag{21}$$

# 6 Two-Layer RL-DRTO Architecture

## 6.1 Purpose and role

The upper layer solves the discrete DRTO problem to compute an economically optimal reactor temperature setpoint trajectory. For online use, we pass a terminal or receding-horizon setpoint to the lower layer. In the RL variant, a trained policy directly maps measurements to safe setpoints without online optimization.

## 6.2 Setpoint handoff

After solving the DRTO over $k = 0..N$, pass

$$T_{R,sp} = T^*_{R,N}. \tag{22}$$

Alternatively, use a policy $\pi_\theta$:

$$T_{R,sp} = \pi_\theta(s_k), \qquad s_k = [T_R, T_J, e_k, F_J, \ldots]. \tag{23}$$

## 6.3 Environment: PI-controlled CSTR (updated dynamics)

The environment represents the plant dynamics; the PI controller tracks $T_{R,sp}$.

$$V_R \frac{dC_R}{dt} = F_R\, C_{R,in} - F_R\, C_R - k(t)\, V_R\, C_R, \tag{24}$$

$$V_R \frac{dT_R}{dt} = F_R\, T_{R,in} - F_R\, T_R - \frac{\Delta H}{\rho C_p}\, k(t)\, V_R\, C_R - \frac{U}{\rho C_p}\, A_R\, (T_R - T_J), \tag{25}$$

$$V_J \frac{dT_J}{dt} = F_J\, T_{J,in} - F_J\, T_J + \frac{U}{\rho_J C_{p,J}}\, A_R\, (T_R - T_J), \tag{26}$$

with $k(t) = k_0 \exp(-E_a/(RT_R))$.

## 6.4 PI controller

Let $e(t) = T_{R,sp} - T_R(t)$. Discrete incremental form:

$$\Delta u_k = K_c \Big[ e_k - e_{k-1} + \frac{\Delta t}{\tau_I} e_k \Big], \tag{27}$$

$$F_J^k = F_J^{k-1} + \Delta u_k, \qquad F_{J,\min} \le F_J^k \le F_{J,\max}. \tag{28}$$

## 6.5 RL observations and reward

$$s_k = [T_R, T_J, e_k, F_J], \tag{29}$$

$$r_k = P_{\text{prod}} F_R (C_{R,in} - C_R) - P_{\text{energy}} U A_R |T_R - T_J| - \lambda_1 (T_R - T_{R,sp})^2. \tag{30}$$

## 6.6 Execution loop

(1) DRTO/RL computes $T_{R,sp}$ every $\Delta t_{\text{RTO}}$.

(2) PI adjusts $F_J$ to track $T_{R,sp}$.

(3) Plant evolves under the continuous dynamics.

(4) Measurements feed back to the DRTO/RL layer.

## 7 Notes on Dynamics Placement

All physical plant dynamics reside in the lower (environment) layer. The upper DRTO layer contains a predictive model used for optimization, here discretized by backward–Euler. An RL policy can replace repeated online optimization by directly mapping observed states to safe setpoints.